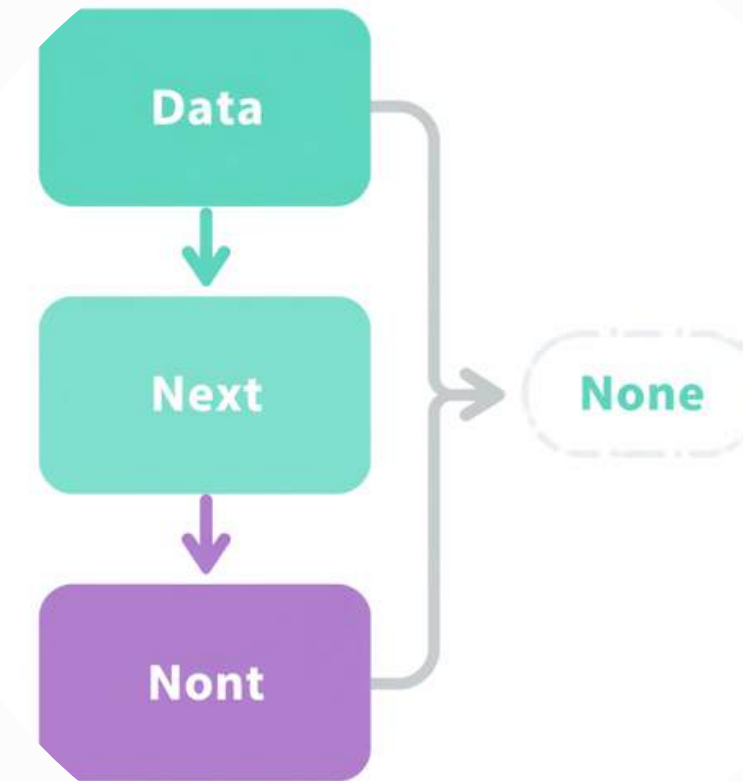




Welcome Aboard!

Building Your First Linked List in Python



Real Train vs Python Linked List

 Engine = Head pointer (start of chain)

 Bogie = Node (data + next pointer)

 Coupling hook = Next pointer

 Last bogie points to None

 ? If Coach B1 only knows who comes after, how does Engine pull Coach B5?

Anatomy of a NodeThe Bogie!



Box 1: DATA

Stores the passengers and luggage — the actual information inside the bogie. In Python: `self.data` holds any value you put in!

Box 2: NEXT

The coupling hook that connects to the next bogie. In Python: `self.next` points to the next Node. Last bogie? `self.next = None`!

Bogie Analogy Recap

Each bogie carries passengers (data) and has a hook (next) to connect to the next bogie in the train chain!



Write Your First Node Class!

```
class Node:  
    def init(self, data):  
        self.data = data  
        self.next = None
```

`self.data` stores what's inside the bogie 🚂
`self.next = None` means no bogie is hooked yet!

✎ Type this in your Planrs Code Editor!



Connecting Two Bogies!

```
bogie1 = Node("Passengers")
```

```
bogie2 = Node("Pantry")
```

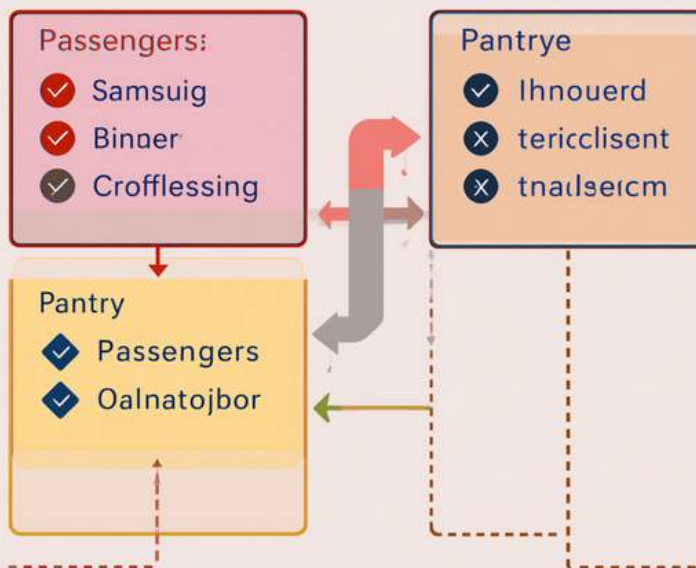
```
bogie1.next = bogie2
```

Now bogie1 points to bogie2 —
just like two hooked coaches! 🚂



Shrt ardly-heae!.

- Set, abpit trestar arst tby grade:
- The rêss ovet tor seusnaus kah pour, and toroitidr grole i spares.



Train Engine

The LinkedList Class!

```
class LinkedList:  
    def init(self):  
        self.head = None
```

🚂 head = None means the train has no bogies yet!
head is the Engine — it starts the whole chain.

When we add bogies, head points to the first Node.
No head = No train. head is where the journey begins!



Adding Bogies with The Append Method!



Step 1 & 2: Create & Check Head

Create `new_node = Node(data)`. If no head exists yet, set `head = new_node`. The new bogie becomes the engine!



Step 3: Walk to the Last Bogie

Use a loop: while `last.next` is not `None`, keep moving forward. Like walking coach by coach to find the last one!



Step 4: Hook the New Bogie!

Set `last.next = new_node` to attach the new bogie at the end. The train grows longer with every `append()`!

Ticket Collector

Walk — Display Method

Like a ticket collector checking each coach one by one, display() visits every bogie from head to None!

```
current = self.head
```

```
while current:
```

```
    print(f'[{current.data}] -> ')
```

```
    current = current.next
```

```
print("None")
```



Your Turn!

Build the **Snack Express!**

Challenge

Task: Add your favourite snacks as bogies!

```
samosa_express = LinkedList()
```

```
samosa_express.append("Samosa")
```

```
samosa_express.append("Jalebi")
```

```
samosa_express.append("Dhokla")
```

```
samosa_express.display()
```

Hint: Use `append()` and then `display()` to see your Snack Express chug along!





Recap & Quiz Time

-  Node = data + pointer (bogie + hook)
-  Head = engine, start of the train
-  Append adds bogies at end
-  Display visits each bogie one by one



Q1: What does self.next store? (a) data (b) Reference to next Node ☒ (c) Engine number

Q2: Next value of last bogie? (a) 0 (b) None ☒ (c) Head

Thank You !